



Name: Cohort/Batch: Date: Score:

QUANTUM COMPUTING PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Emerging Technologies

TOTAL: 10 QUESTIONS

Q1 What is Quantum Computing and what problem in Emerging Technologies does it solve?

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Q6 How does Quantum Computing handle reliability and high availability?

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Q2 What are the core components or concepts in Quantum Computing?

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Q7 What observability does Quantum Computing provide out of the box?

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Q3 How does Quantum Computing compare to its main alternatives?

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Q8 What are common performance bottlenecks in Quantum Computing?

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Q4 How do you get started with Quantum Computing for a new project?

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Q9 What security best practices apply when running Quantum Computing?

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Q5 What configuration options most affect Quantum Computing behavior in production?

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Q10 What mistakes do teams commonly make when adopting Quantum Computing?

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Quantum Computing is a tool or platform

Mitigation

Quantum Computing is a tool or platform that automates or simplifies a specific concern in the Emerging Technologies domain, reducing manual effort and operational complexity for engineering teams.

A6 Quantum Computing provides HA through leader election, replication, or stateless

availability

Quantum Computing provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Quantum Computing has key building blocks that

Primitives

Quantum Computing has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Quantum Computing exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

endpoints

A3 Quantum Computing trades off simplicity against feature richness

hardening

Quantum Computing trades off simplicity against feature richness differently than its alternatives. Choose Quantum Computing when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

performance

A4 Install Quantum Computing via its package manager

Gotchas

Install Quantum Computing via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Quantum Computing with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

Assessment

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

concurrency

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

documentation and